

Semiconductor Etch Process Optimization

A 2³ Full-Factorial Design of Experiments (DOE) Analysis Using JMP

Project Objective: Evaluate the effects of RF Power, chamber Pressure, and Gas Flow on etch rate, identify statistically significant interactions, and determine settings that maximize the predicted response within the experimental design space.

Project Overview

This portfolio project demonstrates an end-to-end DOE workflow in JMP: experimental design, factorial model fitting, effect screening, ANOVA, model-fit assessment, prediction profiling, and desirability-based optimization. The emphasis is not only on statistical significance, but on translating the model into an engineering decision.

Software	JMP Pro 17 (virtual lab)
Design	2 ³ full factorial with 2 replicates
Response	Etch Rate (nm/min)
Data source	Synthetic educational semiconductor-process data

Data-use note: All values are synthetic and are used only to demonstrate DOE methodology. They do not represent production data from any employer or semiconductor manufacturer.

1. Experimental Design

A full factorial design was selected to estimate the main effects of three controllable process factors and their two-factor interactions. Each factor was studied at two levels. The 2^3 design contains eight treatment combinations; each combination was replicated twice, giving 16 total observations.

Factors and Response

Variable	Role	Low	High
RF Power	Factor	200 W	300 W
Pressure	Factor	20 mTorr	40 mTorr
Gas Flow	Factor	40 sccm	60 sccm

Response: Etch Rate (nm/min). The project objective is to maximize the predicted etch rate within the tested factor ranges.

Experimental Dataset

Run	RF Power	Pressure	Gas Flow	Rep	Etch Rate
1	200	20	40	1	82
2	200	20	40	2	84
3	300	20	40	1	104
4	300	20	40	2	107
5	200	40	40	1	75
6	200	40	40	2	78
7	300	40	40	1	91
8	300	40	40	2	94
9	200	20	60	1	91
10	200	20	60	2	93
11	300	20	60	1	119
12	300	20	60	2	117
13	200	40	60	1	80
14	200	40	60	2	83
15	300	40	60	1	101
16	300	40	60	2	104

2. DOE Model Specification

Etch Rate was modeled as a continuous response using Standard Least Squares. The model included the three main effects and all two-factor interactions. Run number and replicate number were not treated as process factors.

Model terms: RF Power + Pressure + Gas Flow + RF Power×Pressure + RF Power×Gas Flow + Pressure×Gas Flow

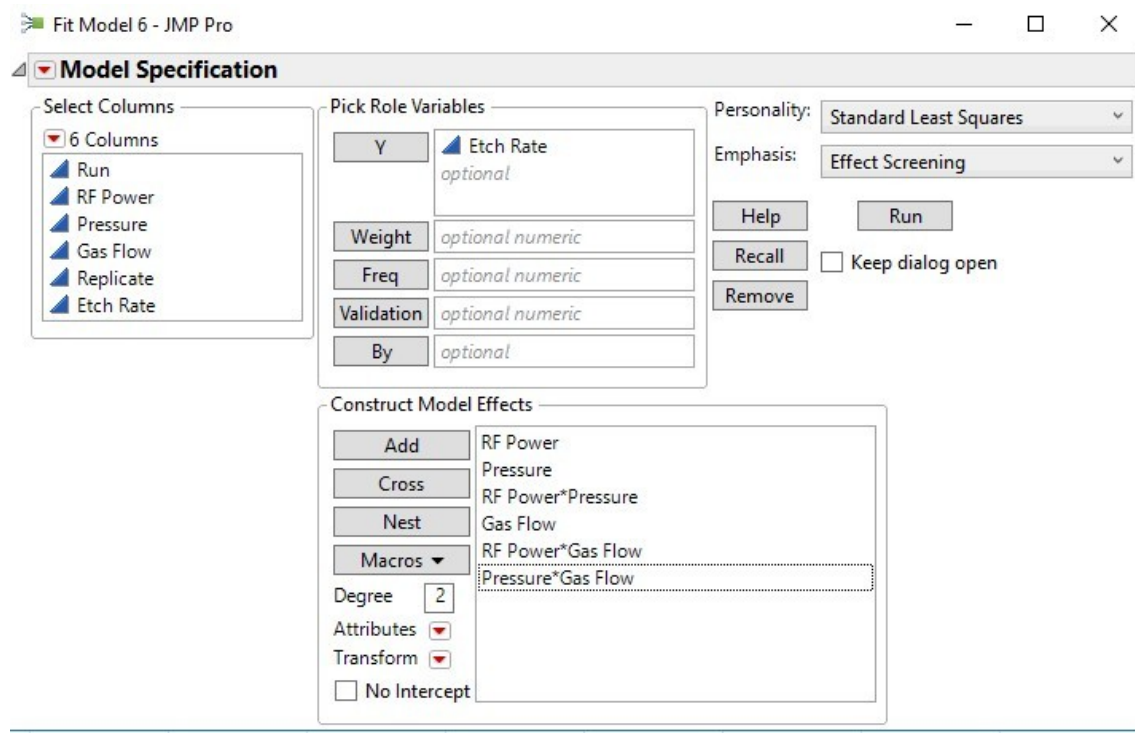


Figure 1. JMP Fit Model specification showing Etch Rate as the response and the six factorial model terms.

Why include interactions?

A main effect describes the average influence of one factor. An interaction asks whether the effect of one factor changes depending on the level of another factor. This is a key advantage of factorial DOE over changing one factor at a time.

Interpretation rule: A significant interaction means the factors should not be interpreted completely independently; the response to one factor depends on the setting of the interacting factor.

3. Effect Summary and Significant Factors

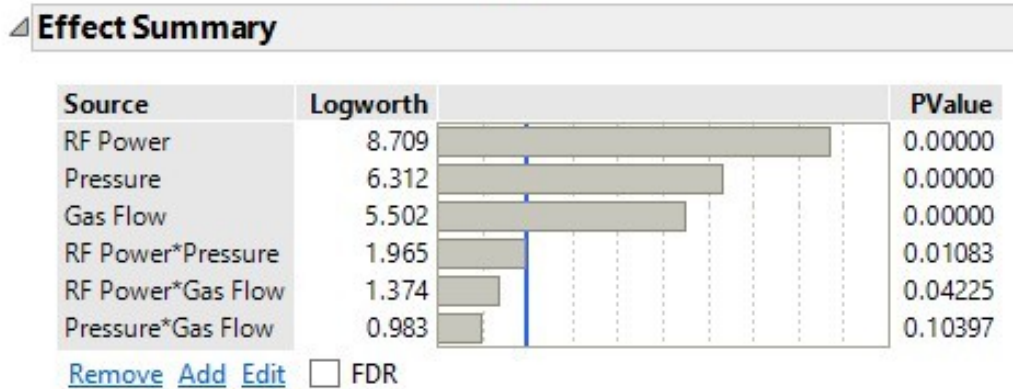


Figure 2. JMP Effect Summary for the fitted factorial model.

The Effect Summary identified RF Power, Pressure, and Gas Flow as statistically significant main effects ($p < 0.001$). Two interactions were also significant: RF Power $\times$ Pressure ($p = 0.0108$) and RF Power $\times$ Gas Flow ($p = 0.0423$). The Pressure $\times$ Gas Flow interaction was not statistically significant at $\alpha = 0.05$ ($p = 0.104$). RF Power had the largest LogWorth and therefore showed the strongest statistical evidence of an effect on etch rate in this model.

Effect	p-value	Conclusion ($\alpha=0.05$)
RF Power	<0.0001	Significant
Pressure	<0.0001	Significant
Gas Flow	<0.0001	Significant
RF Power $\times$ Pressure	0.0108	Significant
RF Power $\times$ Gas Flow	0.0423	Significant
Pressure $\times$ Gas Flow	0.1040	Not significant

Engineering meaning: All three process settings influence the response, and the significant RF Power interactions indicate that the effect of RF Power changes with Pressure and Gas Flow.

4. ANOVA and Model Fit

Summary of Fit				
RSquare		0.989504		
RSquare Adj		0.982507		
Root Mean Square Error		1.796988		
Mean of Response		93.9375		
Observations (or Sum Wgts)		16		
Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	6	2739.8750	456.646	141.4129
Error	9	29.0625	3.229	Prob > F
C. Total	15	2768.9375		<.0001*

Figure 3. Summary of Fit and Analysis of Variance (ANOVA).

ANOVA showed that the overall factorial model was statistically significant ($F = 141.41$, $p < 0.0001$), providing strong evidence that the modeled factors and/or their interactions explain variation in etch rate. The model accounted for 2739.88 of the total 2768.94 sum of squares, while 29.06 remained as unexplained error.

The model explained approximately 98.95% of the observed variation ($R^2 = 0.9895$). The adjusted R^2 remained high at 0.9825 after accounting for model complexity. RMSE was 1.797 nm/min, indicating relatively small residual variation compared with the observed response range.

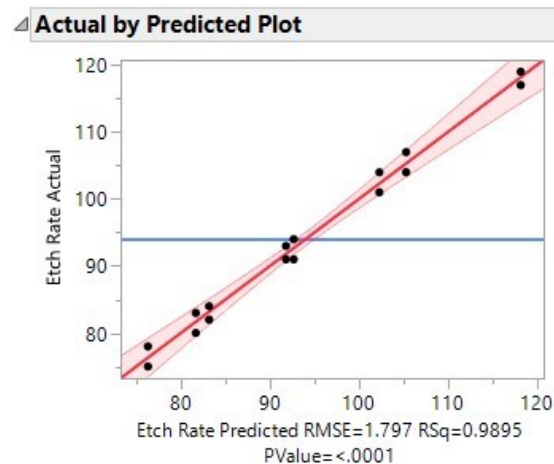


Figure 4. Actual by Predicted plot. Black points are individual observations; the x-axis is model-predicted Etch Rate and the y-axis is actual Etch Rate. Points close to the diagonal indicate good agreement.

Model-fit takeaway: The fitted model is highly significant and closely reproduces the synthetic observations. High R^2 alone is not sufficient; ANOVA, residual behavior, and confirmation runs should also be considered.

5. Prediction Profiler and Process Optimization

The Prediction Profiler translates the fitted statistical model into process-setting decisions. In each factor panel, the red vertical line marks the selected factor setting, while the red horizontal line marks the corresponding predicted response. The sloped fitted lines show how the predicted etch rate changes as one factor changes while the other factors are held at their selected settings.

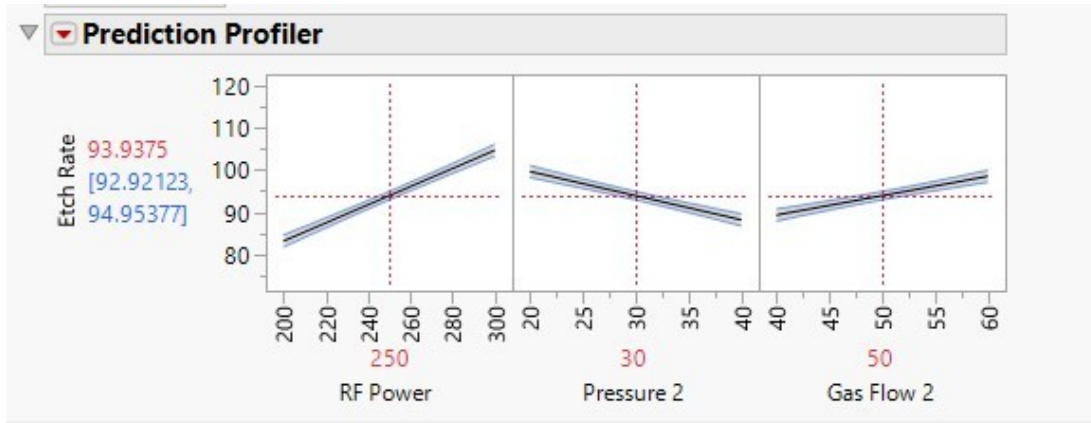


Figure 5. Prediction at the center settings: RF Power = 250 W, Pressure = 30 mTorr, Gas Flow = 50 sccm. Predicted Etch Rate = 93.94 nm/min.

At the center settings, RF Power and Gas Flow show positive slopes, while Pressure shows a negative slope. Because the model contains significant interactions, these slopes are conditional on the current settings of the other factors and should not be interpreted as universal process relationships.

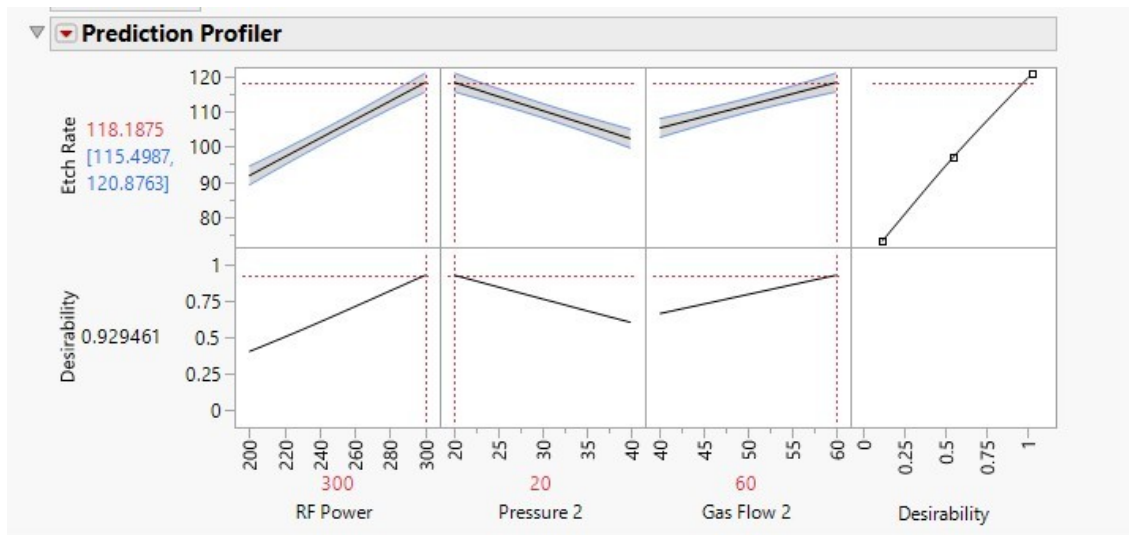


Figure 6. Desirability optimization. JMP selected RF Power = 300 W, Pressure = 20 mTorr, and Gas Flow = 60 sccm, with a predicted Etch Rate of 118.19 nm/min and desirability of 0.929.

Optimized settings: RF Power = 300 W | Pressure = 20 mTorr | Gas Flow = 60 sccm | Predicted Etch Rate = 118.19 nm/min | 95% displayed interval $\approx$ 115.50-120.88 nm/min | Desirability = 0.929

6. Engineering Conclusions and Next Steps

Key Findings

- RF Power, Pressure, and Gas Flow were statistically significant main effects on Etch Rate ($p < 0.001$).
- RF Power $\times$ Pressure ($p = 0.0108$) and RF Power $\times$ Gas Flow ($p = 0.0423$) were significant interactions; Pressure $\times$ Gas Flow was not significant at $\alpha = 0.05$.
- The overall factorial model was statistically significant ($F = 141.41$, $p < 0.0001$) and explained 98.95% of the observed variation ($R^2 = 0.9895$).
- Within the tested design space, desirability optimization selected 300 W RF Power, 20 mTorr Pressure, and 60 sccm Gas Flow, predicting an Etch Rate of approximately 118.19 nm/min.

Engineering Interpretation

The analysis demonstrates why factorial DOE is valuable for process development: it identifies both individual factor effects and interactions that would be difficult to recognize using one-factor-at-a-time experimentation. The significant RF Power interactions show that process settings should be evaluated as a system rather than optimized independently.

Recommended Next Steps

- Run independent confirmation experiments at or near the predicted optimum to verify the model prediction.
- Evaluate additional responses such as uniformity, defect rate, selectivity, surface quality, or throughput so optimization is not based on etch rate alone.
- If curvature is suspected near the optimum, extend the study with center points or a response-surface design.
- For a real manufacturing application, define engineering guard bands and confirm process robustness before implementation.

Portfolio statement: This project demonstrates practical use of JMP for factorial DOE, interaction analysis, ANOVA, model validation, prediction profiling, and engineering optimization using synthetic semiconductor-process data.